



Orthopaedics WORKBOOK



Dr. Akshay Saxena



INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.



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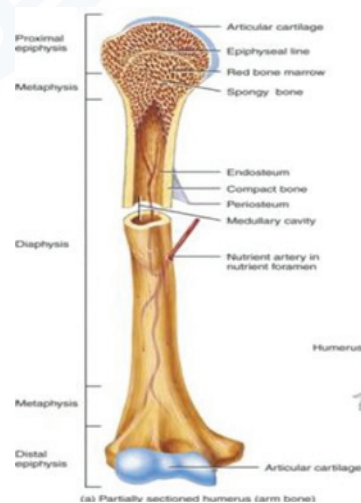
GENERAL ORTHOPAEDICS AND POLY TRAUMA

Orthopaedics-

- ⇒ Ortho - Straight
- ⇒ Paedics - child

□ Anatomy of Long Bone

1. Epiphysis - Ends of the bone.
2. Metaphysis - Bone part between Epiphysis and Diaphysis
3. Diaphysis - Shaft of the Bone (Strongest)



4. Growth plate/physis:- Cartilage zone which

(Between Epiphysis and metaphysis) Contributes to longitudinal

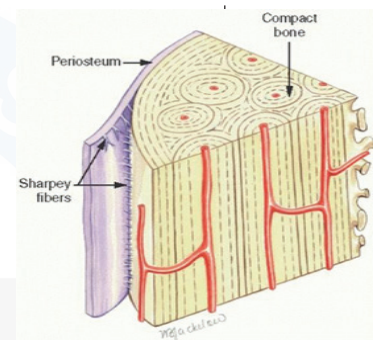
Growth of bone

or

5. Periosteum - Fibrous layer covering of the bone

- Inner (Cambium) layer increases thickness of the bone or

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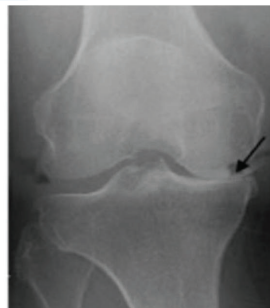
6. Sharpey's Fibres - Attach Periosteum to Bone

7. Articular Cartilage - Hyaline Cartilage

Osteoarthritis - Age Related

Degeneration of Articular Cartilage

M/C - Knee Joint (Medial compartment)



Structure of Bone

□ Types of Bone :

Immature (Woven)

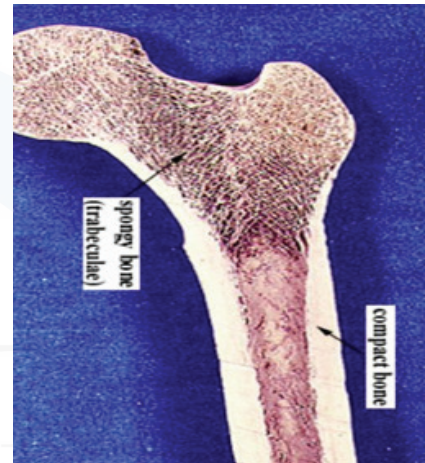
Bone

I. Embryonic life

II. Initial stages of (#)

Fracture healing.

Mature Bone (Lamellar)



Cancellous/Spongy Bone

i. Present in

.....

ii. Vascularity

iii. Soft bone

iv. Basic Unit

v. High chances of

.....

Compact or cortical bone

i. Present in

.....

ii. Vascularity

iii. Hard bone

iv. Basic Unit

v. High chances of

.....

Composition of Bone (Bone Cells + Extra-Cellular Matrix)

Bone Cells :

3 Types :

1. Osteoblast :

- i. Bone Forming cells
- ii. Also synthesise
- iii. Rich in enzyme
- iv. Also controls Osteoclastic activity
(Has Receptors for)

2. Osteocytes :

- i. Mature or spent
- ii. Rich in

3. Osteoclasts :

- i. Cells for
- ii. Derived from
- iii. Rich in enzyme

Clinical Examples :

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Active space



Extra-Cellular Matrix :

2 types :

1. Organic Matrix :

.....

2. Inorganic Matrix :

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Clinical Example :

A condition in which child has defect in

Type 1 collagen since birth ??

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Fracture

Partial or complete Break in Bone

M/C Sign:-

Other Signs:-

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Fracture Healing

◆ Stages In Fracture Healing of Cortical Bone: (Frost)

1. Stage of Haematoma formation (7 days)

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2. Stage of inflammation / Granulation tissue formation (7 days-3wks)

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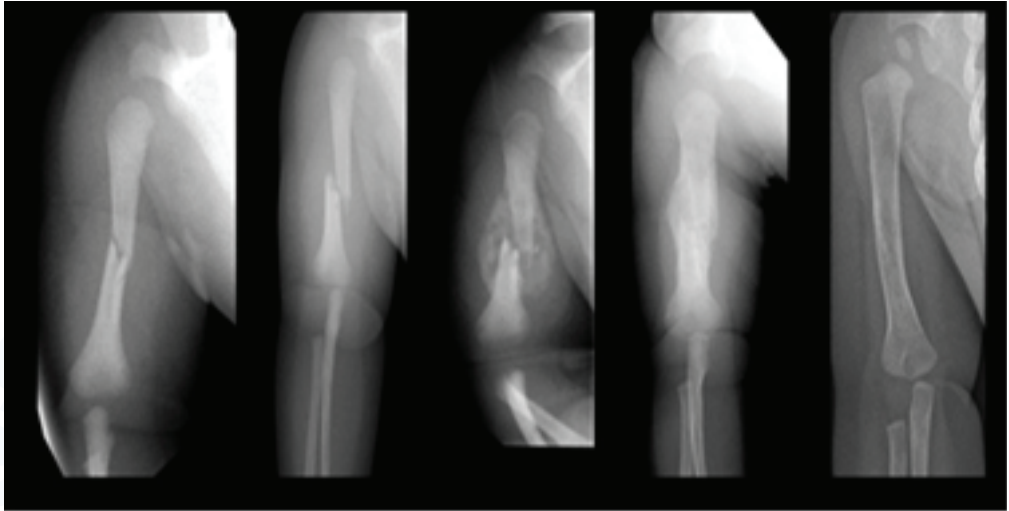
3. Stage of Callus Formation (3 weeks - 12 weeks)

- i. Callus is
- ii. First stage which is
- iii. First stage in which # becomes
- iv. First appears at the end of after injury

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4. Stage of Remodelling (3 months - 12 months)

i. Conversion of Callus into



Classification of Fractures

A. On the basis of Aetiology (Cause) :

1. Traumatic



2. Pathological

i. Fracture in an underlying

iii. C/F : Pain is present before fracture occurs



iv. M/C Cause :

v. Other Causes :

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vi. Treatment

Stabilization of bones :-

Internal Fixation

- Surgical stabilization of bones with implants which Lie inside Patient's body

I. Plate fixation : -

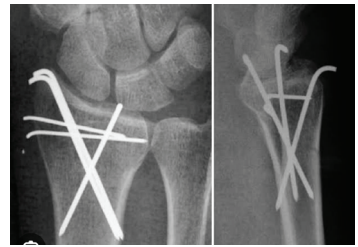


II. Intra medullary nailing



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III. Screw / k-wire fixation



External Fixation :-

Surgical Stabilization of bones with implant outside patient's body

Ex:-



B. Relationship with External Environment :

1. Closed/Simple

- No External wound overlying #
- # Does not communicate with external environment

2. Open/Compound

- External wound overlying # is present
- M/c Site :
- Higher chances of and
- Classified by



Treatment :

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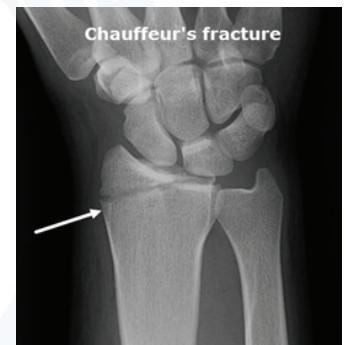
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Active space

MIST

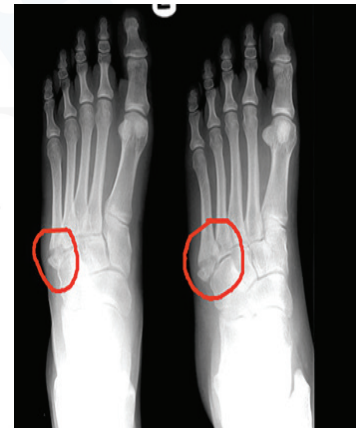
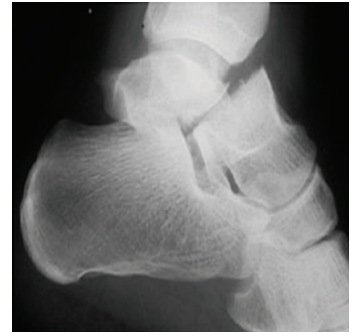
Fracture Eponyms

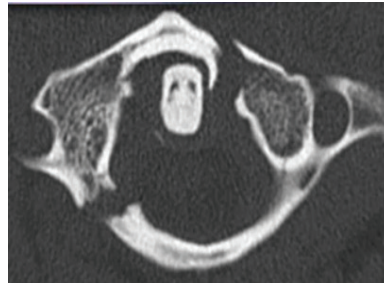




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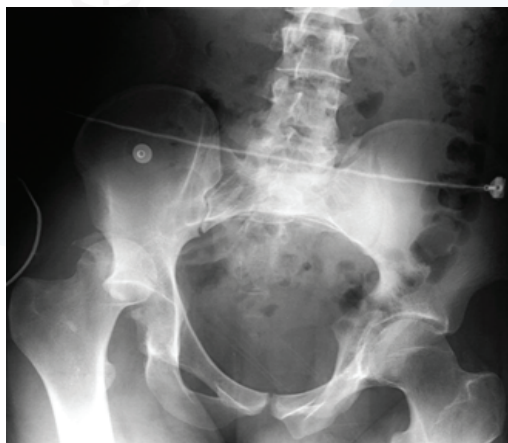
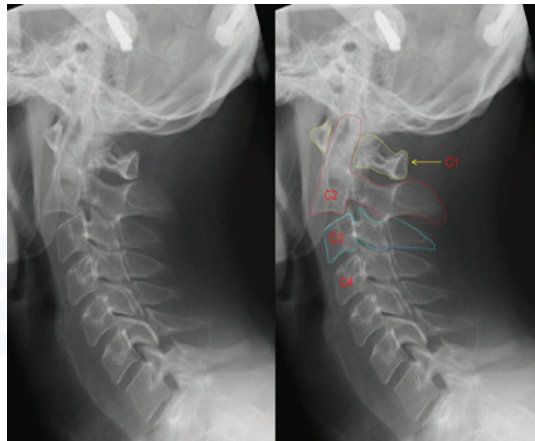
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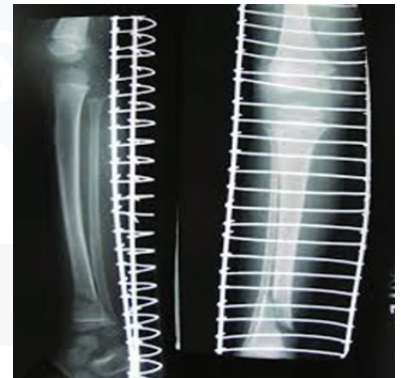
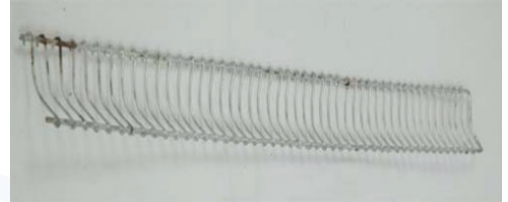
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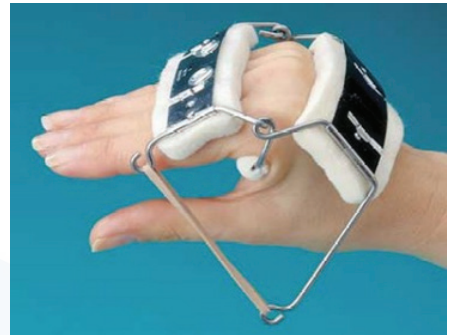
Splints in Orthopaedics



Active space



Active space



Active space



Active space

Important Traction



Active space

Complications of Fractures

- 1 Immediate Complications:(At the time of hospital presentation)
 - Hypovolemic shock
 - Injury to the nerves and blood vessels.
- 2 Early Complications : (24 to 72 hrs.)
 - Acute respiratory distress syndrome (ARDS)
 - Fat embolism syndrome
 - Deep venous thrombosis & pulmonary embolism
 - CRUSH syndrome
 - Compartment syndrome
- 3 Late Complication: (Weeks to months after the injury)
 - Delayed union, non-union & malunion
 - Shortening, Deformity, Stiffness.
 - Avascular necrosis
 - Myositis ossificans
 - Reflex sympathetic dystrophy. (RSD)

Hypovolaemic Shock:

Hypovolemia → Hypoperfusion to vital organs

Causes :

- # Shaft of femur → 1000 - 1500 ml of blood loss
- # Pelvis (open book fracture) → 1500 - 2000ml blood loss

C/f :

- i. Blood Pressure
- ii. Pulse Rate
- iii. Urine Output

Treatment

1. Crystalloids- (Isotonic Solutions.....)
2. Best Replacement
3. Colloids:.....
4. Best prognostic marker

Acute Respiratory Distress Syndrome

Polytrauma (> 2 organ systems involved in trauma)



↑ Release of Inflammatory mediators



Damage to small blood vessels of lungs



Fluid comes out in the lung parenchyma



Impairs gas exchange in Lungs



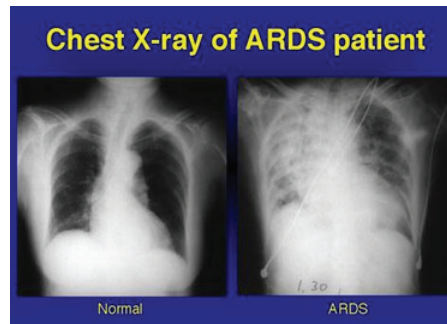
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C/f :-

- i. Respiratory Rate
- ii. Partial pressure of O₂

X-Ray:-

..... Pulmonary Infiltrates



Treatment

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Fat Embolism Syndrome

1. Associated with fractures of Diaphysis of Long bones
2. M/C Cause :
3. Pathogenesis :

Long Bone fracture



Release of Bone Marrow in Circulation



Formation of free fatty acids



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Clinical Triad :

- (i)
 (ii)
 (iii)



X-Ray of chest :-

.....



Treatment :-

1. Early # stabilization
2. Low Molecular Weight Heparin
3. Ventilation
4. Steroids

Deep Vein Thrombosis

Prolonged Immobilization (Polytrauma, Spinal Cord Injuries, Post surgery-ex post arthroplasty)



Venous Stasis



Thrombus Formation

Active space

C/f :-

- (i) Swelling
- (ii) Tenderness
- (iii) Skin Pigmentation
- (iv) Can lead to Pulmonary Embolism
in 4-5 days



Swelling



Skin Changes

Clinical Test :

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Investigation :

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Treatment :-

- (i) Prophylaxis DVT Stockings & DVT Pumps
- (ii) LMWH
- (iii) Factor Xa Inhibitors

.....

Crush Syndrome / Traumatic Rhabdomyolysis:

Causes: (1) Earth Quakes (2) Mining Accidents

(3) M/c Cause :-

Pathogenesis :

Crushing of Muscles



Release of MYOGLOBIN

Active space

↓
Obstruction of Renal Tubules

↓
Acute Tubular Necrosis

C/F :

1. Electrolyte Imbalance
2. Hyperkalemia + Hypocalcemia + Metabolic Acidosis
3. Renal Symptoms

Treatment :

1. I/v Fluid & Electrolyte Management
2. Alkalinization of Urine : -
3. Haemodialysis

Compartment Syndrome:

Increase of pressure inside the anatomical compartment causing compression of :

1.
2.

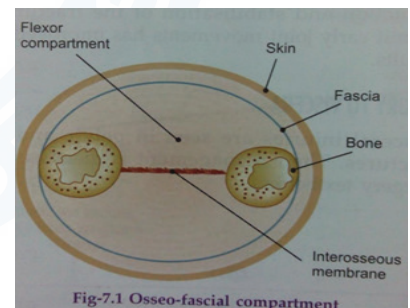
C/f :- Pain (not responding analgesics)

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Causes:

1. Upper Limb

- i. M/C compartment involved is Deep Anterior compartment of forearm
- ii. Muscles involved
- iii. In Children
- iv. In Adults



2. Lower Limb

- i. M/ C Cause
- ii. M/C Compartment Involved

Clinical Test :-

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Treatment:

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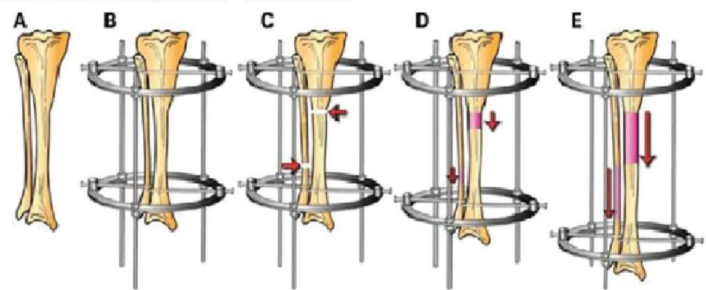
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Late Complications

1. Shortening

- < 2 cm :
- > 2 cm :



Avascular Necrosis

Definition : Death of bone tissue in absence of adequate blood.

Most common cause

M/C Site

Other causes :

1. Trauma
2. Drug induced: Steroids
3. Alcohol abuse
4. Haemoglobinopathies: sickle cell anemia
5. Storage Disorders : Gaucher's Disease

Traumatic AVN :

Cause / Injury :

Site of AVN :

1. Posterior Hip Dislocation
Fracture Neck Femur

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2. Fracture Waist of Scaphoid :

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3. Fracture Neck Talus :

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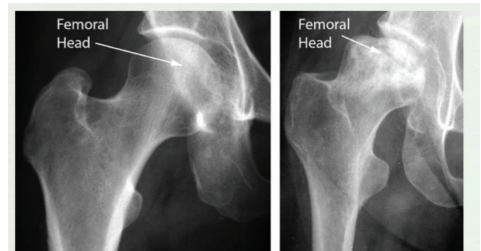
C/F :

1. Pain in Hip alongwith restriction of

X Ray :

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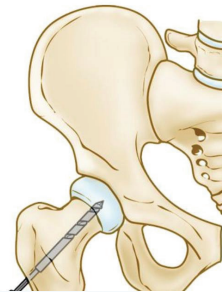


Investigation of Choice :

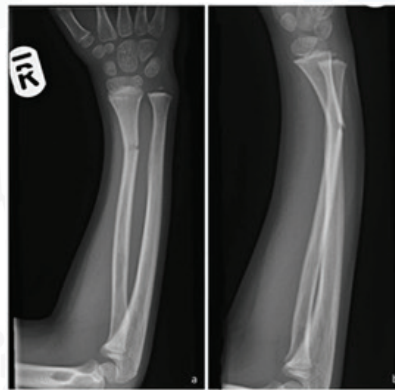
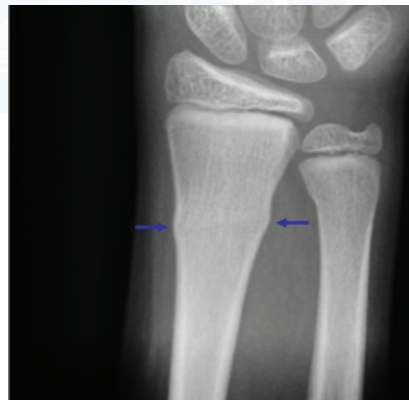
Treatment :

1. Early AVN :

2. Late AVN :



Fractures In Children

1. Incomplete # in children (Break of one cortex and bent of other :
.....2. Bulging/Buckling of cortex in children :
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Active space

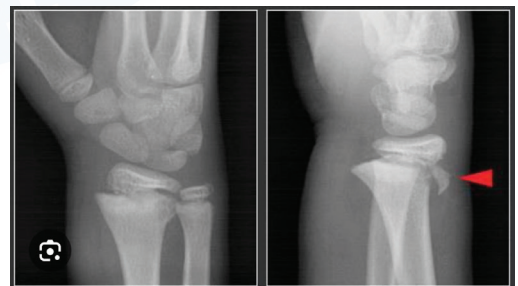
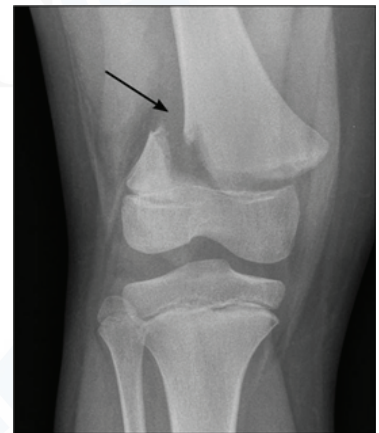
Epiphyseal Injuries

1. Injuries (Fractures) involving the growth plate (physis) in children
2. Classified by " Salter & Harris " classification

Type - I :



Type - II :



Type - III :



III

Type - IV :



IV

Type - V :



V

Active space

Type I & Type II - Good Prognosis

Stress Fracture

Definition : Fracture developing due to repeated loading of bone

Common Sites :

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March Fracture :

Stress fracture of

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Commonly seen in

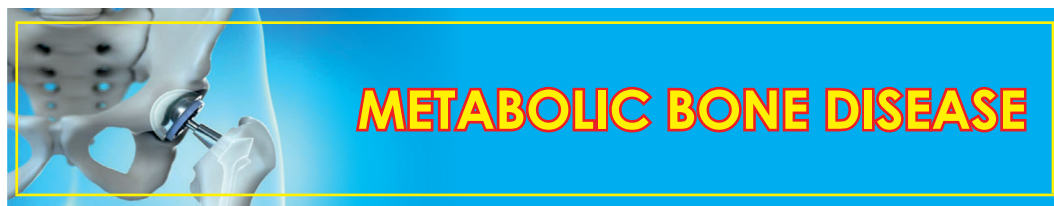
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IOC :- (i) For U/L # :-

(ii) For B/L # :-

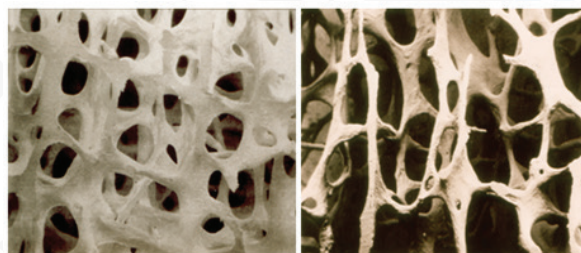


Osteoporosis

M/c metabolic bone disease

M/c cause of pathological fracture

Definition :



Healthy bone

Osteoporotic bone

Causes:

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C/F :

1. Earliest Symptom :
2. Fragility # :
3. Dowager's Hump : Kyphosis

X-Ray :

1. Decrease in Density of bones :
(Ground Glass Appearance)
2. Biconcave/Cod fish/Fish mouth vertebrae



Biochemistry

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DEXA SCAN (Dual energy X-ray Absorptiometry scan)

TScore : Comparison of bone density of patient with 30 yrs old patient of same sex and same race

WHO Classification :

T/T:

1. Inhibit Bone Resorption :

i. Bisphosphonates :

.....



ii. Hormone Replacement Therapy

iii. RANK Ligand Inhibitors :

.....

2). Stimulate Bone Formation:

i. Calcium and Vitamin D supplements

ii.

.....

Frax Scoring

- | | |
|--|---------------------------------|
| 1. Age | 2. Sex |
| 3. Weight | 4. Height |
| 5. Previous H/o of # | 6. Previous H/o HIP# In Parents |
| 7. Rheumatoid Arthritis | 8. H/o Glucocorticoid |
| 9. Secondary Osteoporosis | 10. Smoking |
| 11. H/o Alcohol Intake
(> 3 units or 90 mL per day) | 12. Femoral Neck BMD |

Rickets & Osteomalacia

Definition : Defect in mineralization of bone

- In Children - Rickets
- In Adults - Osteomalacia

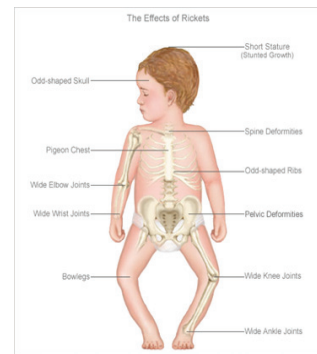
Rickets : Defect in mineralization at growth plate

Most Common Type :

C/F :

1. Skull Findings : (Earliest)

- i. Softening of bone :
- ii. Prominent frontal bone :
- iii. Squarring of skull :

**2. Chest Findings :**

- i. Prominence of chest :
- ii. Costochondral swellings :
- iii. Depression / Groove due to pull of Diaphragm :

**3. Knee Findings :**

- i. Genu varus
 - M/C Cause in children :
 - M/C Cause in Adults :
- ii. Genu Valgus :
 - M/C Cause in children :
 - M/C Cause in Adults :

**iii. Windswept Deformity : (Genu Varus in one knee and Genu Valgus in other knee)**

- M/C Cause in children :
- M/C Cause in Adults :

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X-Rays:

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**Biochemistry: (Nutritional Rickets)**

- Serum calcium
- Serum phosphorus
- Serum U it D
- Serum PTH
- Serum Alkaline Phosphatase

Treatment of Nutritional Rickets (Stoss Therapy) :

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Osteomalacia

1. More common in
2. Due to deficiency of :

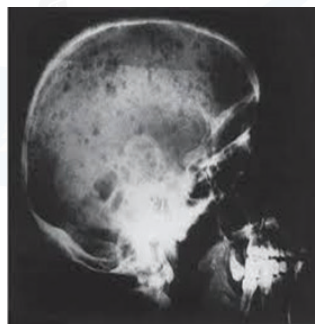
C/F :

- (i) Bone Pains
- (ii) Tiredness / Fatigue & Lethargy
- (iii) Pathological fracture



Primary Hyperparathyroidism

1. Caused due to a
2. High levels of PTH causes :



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Biochemistry

S. Calcium -

S. Po43 -

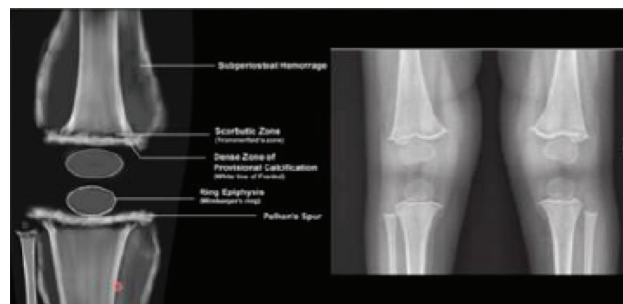
S. ALP -

S. PTH -

Scurvy :

- ✦ Vitamin C deficiency → failure of hydroxylation of Proline and Lysine —
- ✦ C/F : Failure to Thrive
 - Bleeding Gums
 - Subperiosteal Haemorrhage in Distal Femur / Tibia — Pain —
 - Impaired Wound healing

Radiology :



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- ✦ Pencil thin cortex
- ✦ Opaque outline of Epiphyseal ossification centres -
-
- ✦ Dense zone of Provisional Calcification
- ✦ Radiolucent Metaphyseal band under frankel's line -
-
- ✦ Lateral Metaphyseal Spurs -

Osteopetrosis / Marble Bone Disease / Albers Schonberg Disease :

- Defect in Osteoclast function (Resorption)

C/f :

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XRays:

- Bone within Bone appearance

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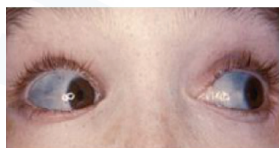


- T/t :

Osteogenesis Imperfecta/Brittle Bone Disease

- Defective Type 1 Collagen formation
- Associated with :

.....
.....
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XRays:

-
-



T/t:

-
-

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Paget's Disease (Osteitis Deformans)

- Disorganized Bone Remodelling — Excessive Osteoclastic Resorption f/b disorganized New Bone formation (Highly vascular but weak immature/woven Bone)

- are defective

- More Common in Males

- Sites :

- C/F : Pain is M/C symptom

Limbs deformed, skin warm to touch

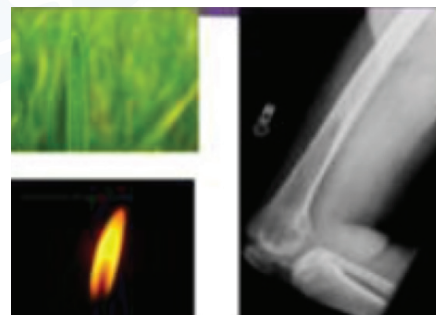
- Biochemistry : S. Calcium —

S. Phosphate —

S. ALP —

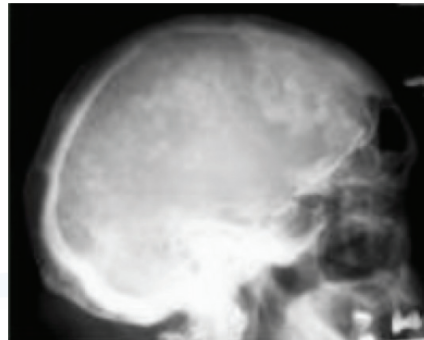
- Long Bone :

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- Skull :

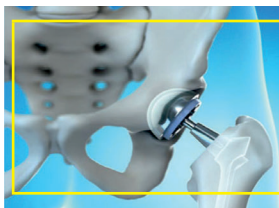
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- Vertebrae :

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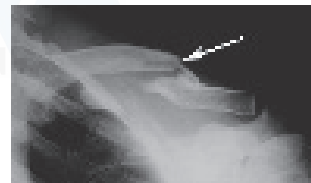
UPPER LIMB TRAUMA

UPPER LIMB TRAUMA

Upper Limb Trauma

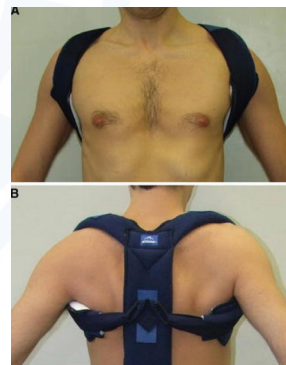
Clavicle: -

- m/c # in
- m/c # in
- m/c site



Rx : -

Conservative : Clavicular Brace / Figure of 8 bandage



Surgery : ORIF (Open Reduction and Internal fixation)

.....

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M/c Complication :

Shoulder Dislocation: -

→ M/c joint to dislocate in body.

→ Types :

Anterior shoulder dislocation (Most common type)

Posterior shoulder dislocation

Inferior shoulder dislocation

Anterior Shoulder Dislocation

1. M/c Type

2. Attitude :-



Clinical Test :

i. Duga's Test :

.....

ii. Hamilton Ruler Test

iii. Calloway's Test

Treatment: - Closed Reduction



Kocher's Technique: (TEAM)

T-

E -

A -

M -

Posterior Shoulder Dislocation

(Head of humerus is posterior to Glenoid after dislocation)

M/C type of shoulder dislocation

Causes :

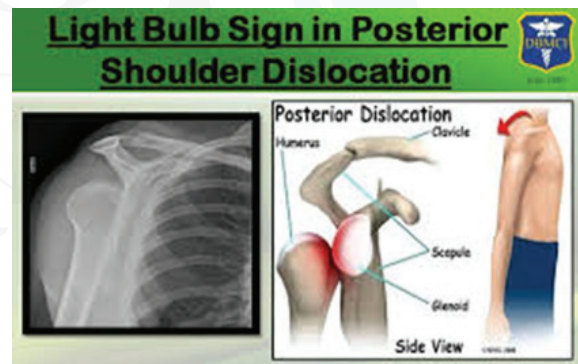
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Attitude :

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X-Ray:

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Treatment

Closed Reduction



If Fails Than Sx

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Inferior Shoulder Dislocation

1. Also called as
2. Attitude :



Recurrent Dislocation

Definition:



3 Or More Dislocation of The Same Joint

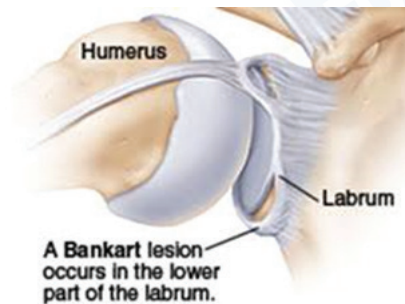
M/C Type :

Pathological Lesions:

- 1 Bankart's Lesion : (90% of Cases)

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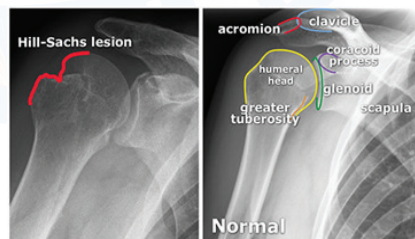


IOC - MRI

2. Hill - Sach's Lesion :

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3. Bony Bankart's Lesion :

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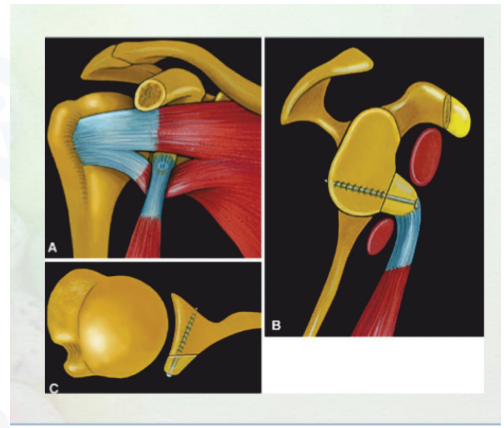


Active space

MIST

Treatment :

1. T/t of choice
2.
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Fracture Humerus

1. Fracture Proximal Humerus :

- i. Common in Elderly osteoporotic patients
- ii. M/C Nerve Injured :



2. Fracture Shaft of Humerus :

- i. M/C Nerve Injured :
- ii. Holstein - Lewis # :



Treatment of humerus shaft fracture :-

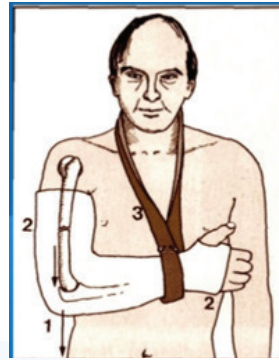
1. Conservative

.....

.....

.....

.....



Non-Union

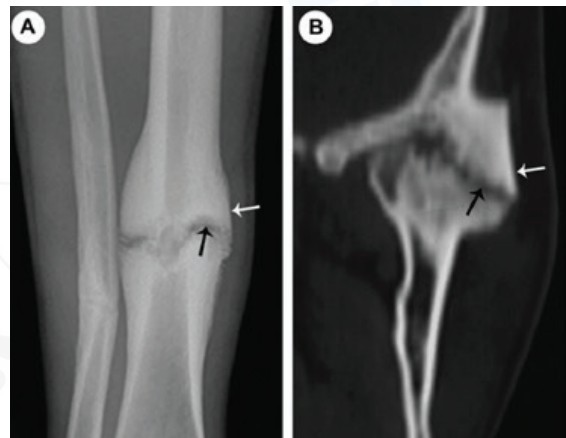
Definition :

.....

.....

Sites of Non - Union :

- i. Neck femur fracture
- ii. Sacphoid fracture
- iii. Lateral condyle humerus



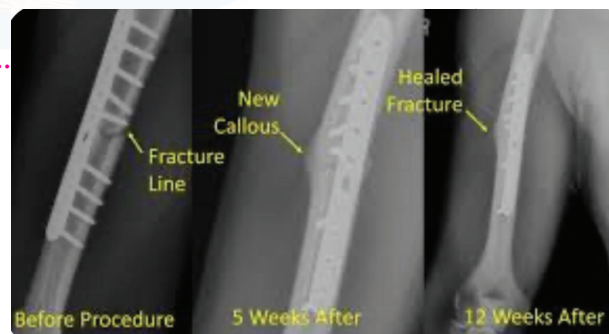
Treatment :

i. ORIF +

ii. M/c site of extracting Bone graft :

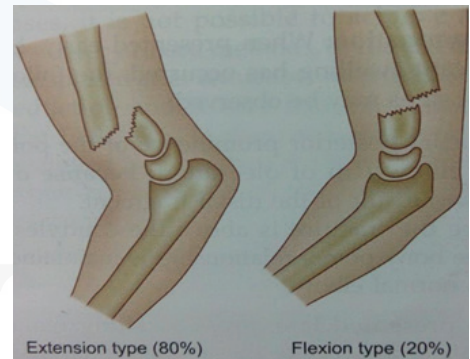
.....

.....



Supracondylar Fracture

1. Definition : # in the distal humerus Metaphysis above condyle
2. Age Incidence :
3. It is Extra - Articular Fracture. :
4. M/ C fracture associated with :
5. Types:



6. Displacements:



7. Diagnosis :

- M/C Blood Vessel Injured :
- M/C Nerve Injured :

8. Treatment :

- i. Undisplaced Fracture : Above Elbow POP

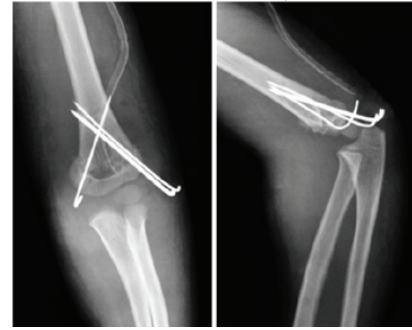
ii. Displaced Fracture :

Closed Reduction / Open Reduction

+

Internal Fixation With Kirschner

Wires (K- Wires)



Complications:

1. Early Complications: (Within 2-3 days)

.....

.....

2. Late Complications:

i. Malunion :

.....

.....

.....

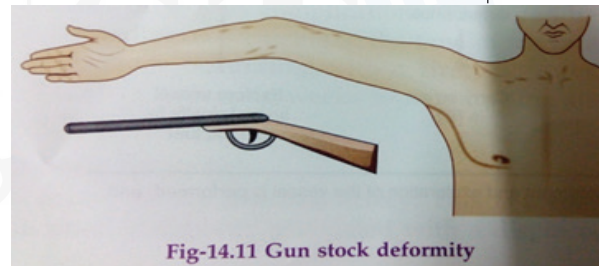


Fig-14.11 Gun stock deformity

ii. Myositis Ossificans :



- Calcification of haematoma within muscle
- M/C Due to
- M/C muscle involved
- M/C after



- iii Volkmann's Ischemic Contracture
Ischaemic / Dead muscles are
replaced by fibrous tissue



Splint used :

Turn Buckle Splint

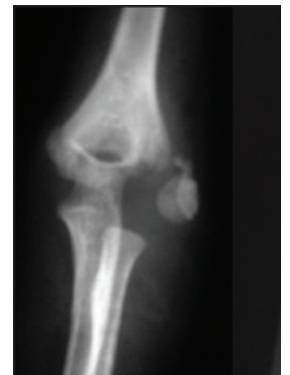
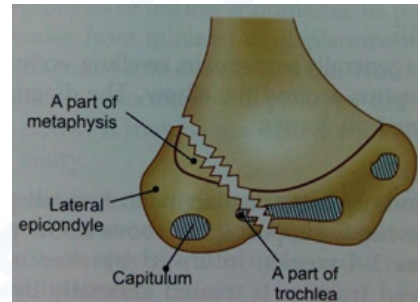


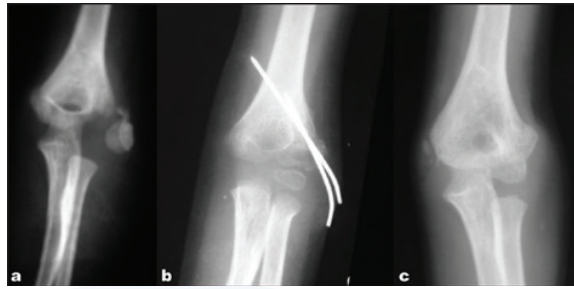
Fracture Lateral Condyle Humerus

1. Lateral condyle fragment :
.....
2. It is Intra - Articular fracture
3. It is
Salter & Harris Epiphyseal Injury
4. Treatment :



OR + IF with K - wires
(Fracture of Necessity)



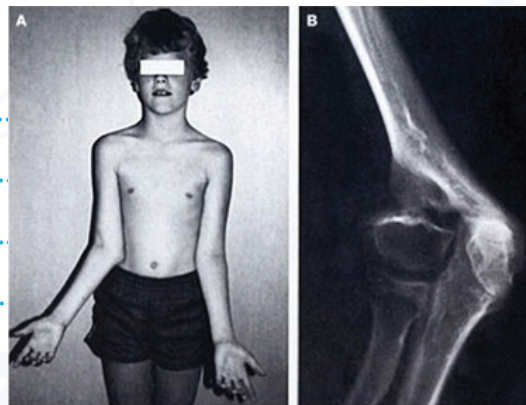


Other fractures of Necessity :

- (i) Fracture Neck of Femur
- (ii) Monteggia fracture Dislocation
- (iii) Galeazzi Fracture Dislocation

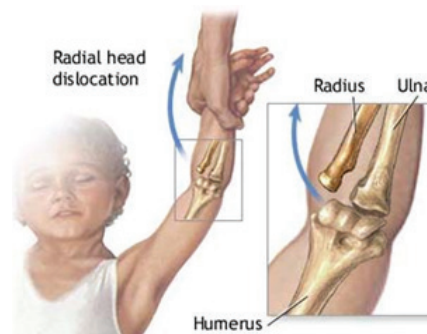
Complications:

- (i)
- (ii)
-
- (iii)



Pulled Elbow (Nursemaid's Elbow)

- 2-5yrs Of Age
- D/T Sudden Pull :
- Radial head comes out of its annular ligament



Active space

C/F :-

1. Pain
2. Restriction of Elbow movements
3. Forearm in attitude of

X Rays : **Normal**

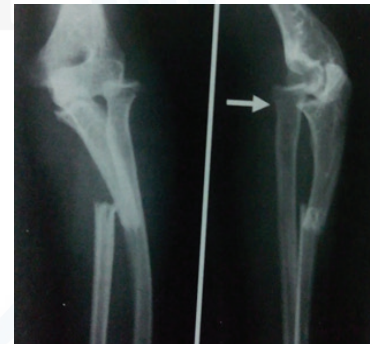
T/t : Closed Reduction by :

.....

Fractures of Forearm and Wrist

of Proximal third of ulna + Radial Head Dislocation :

.....



Distal Shaft of Radius + Inferior Radio-Ulnar joint dislocation :

.....



Colle's Fracture

1. Definition :

Extra - Articular # In
Distal radius metaphysis.

2. Age incidence :

.....
.....

3. It is an Extra - Articular fracture

4. Displacements:

5. Treatment:

(A) Below elbow pop cast with wrist in :-

(i)

(ii)

(B) Thumb is spared in cast

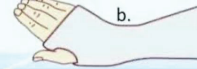
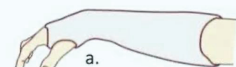
.....
.....
.....

6. Complications :

i. M/C Complication :



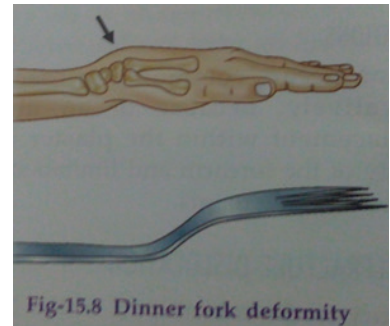
- Colles' Cast is used to immobilize undisplaced or reduced fracture.
- It is below elbow cast with
 - a. Palmar Flexion
 - b. Ulnar Deviation



Active space

ii. Malunion :

iii. Carpal Tunnel Syndrome



iv. Sudeck's Osteodystrophy

(Reflex Sympathetic Dystrophy /Complex Regional Pain Syndrome)

2 types :

Type - 1

i. Caused after soft tissue / bony injury.

ii. M/c Cause :

Type - 2

i. Caused after peripheral nerve injury.

ii. M/c Cause :



C/F: Pain, Swelling, Stiffness of
Involved Extremity

X-Ray:



Treatment:

- Physio & NSAIDS
- Sympathetic Blockade
- Sx- Sympathectomy



Scaphoid Fracture

1. M/C Carpal Bone fracture
2. Age Incidence :-
3. M/C Site

.....

4. C/F : Tenderness in



Investigation :

For Acute # :-

For occult # :-

Treatment :

1. Undisplaced # : Below Elbow POP Cast

⇒ With wrist in

(i)

(ii)

⇒ Thumb is involved in cast



2. Displaced # : Open Reduction + Internal

Fixation with

.....



Complications:

1. M/C
2.
.....
3.

**Injuries of 1st CMC Joint :**

1. Intra articular # of 1st Metacarpal base + subluxation / dislocation of 1st CMC joint:-
.....
.....



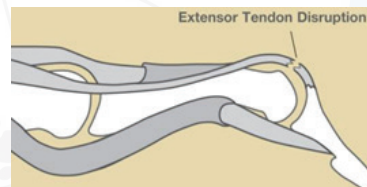
2. # 1st Metacarpal
Base :-
.....
.....



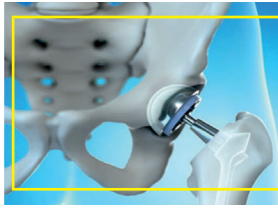
2. # Neck of 5th Metacarpal



Mallet Finger/ Baseball Finger



Active space



LOWER LIMB TRAUMA

LOWER LIMB TRAUMA

Dislocations of Hip

Types:

1. Posterior:

- M/C Type
- Also called as



C/f : (i) Attitude of Injured Limb is

.....

(ii)

X Rays :



Treatment:

Closed Reduction under GA/ Sedation

.....

Complication:

.....

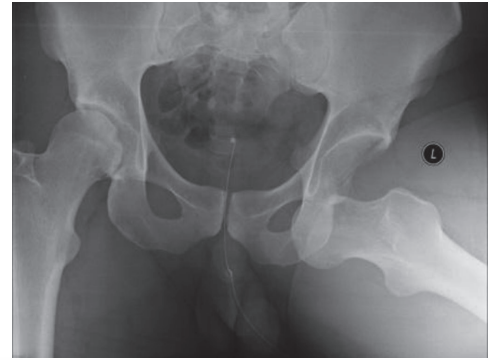
.....

2. Anterior Dislocation :

C/f : (i) Attitude of Injured Limb is

.....

(ii)



3. Central Fracture Dislocation:

.....

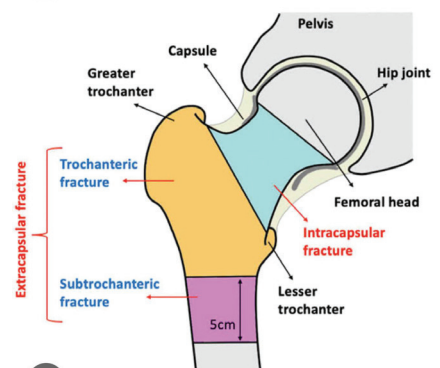
.....



Fractures Around Hip

Intracapsular Vs Extracapsular Fracture:

- 1 # Neck of Femur (Intracapsular #)
- 2 Intertrochanteric #
(Extracapsular#)



MIST

	# Neck Femur	Intertrochanteric fracture :
Location	Intracapsular	Extracapsular
Sex Incidence :		
Velocity of Trauma :		
Symptoms : (Pain, Swelling & Skin discolouration / Bruising)		
Attitude of Injured Limb : (External Rotation)		
Prognosis :		

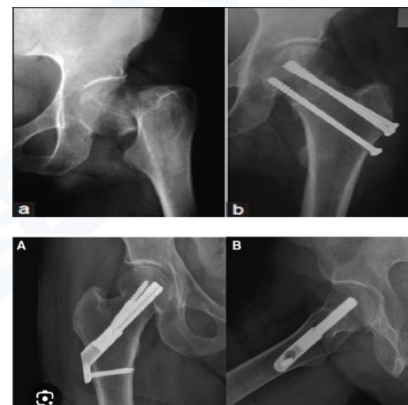
Acute Fracture Management of fracture Neck Femur :

(Head Preservation In Young And Head Replacement In Elderly)

1 < 60 Yrs :

Goal :

Treatment :



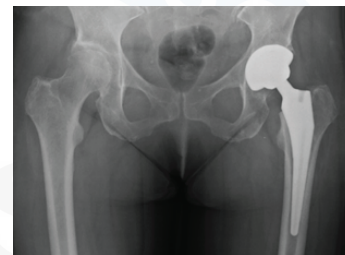
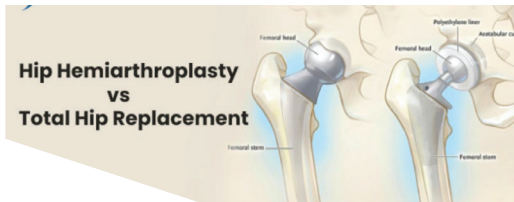
2 > 60 Yrs :

Goal :

Treatment :

Active space

Hemiarthroplasty vs Total Hip Arthroplasty :



Complications:

1.
2.



Intertrochanteric Fracture

1. Fracture extending from Greater trochanter to lesser trochanter
2. Extracapsular #



Active space

3. Fracture in the proximal femoral Metaphysis

Femoral Neck - Shaft Angle:

(i) Normal Neck-Shaft angle :

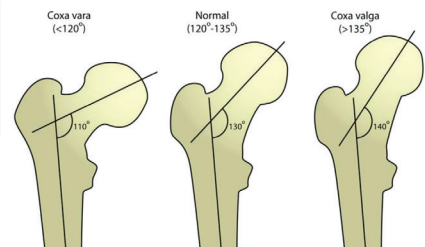
.....

(ii) If angle is < 120 deg :

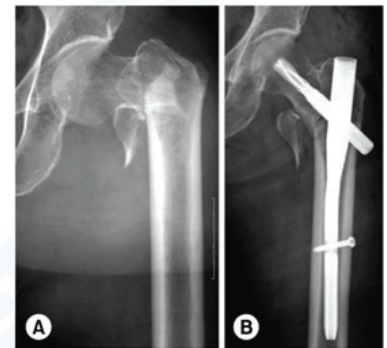
.....

(iii) If angle is > 135 deg :

.....



Treatment:



Complications:

.....



Fracture Shaft Femur

1. High Velocity Injury
2. Seen in Young Patients

Treatment :

1. Splint Used :
Thomas Splint

Rx: Depend on Age:

< 2 yrs :

.....

2-5 yrs :

.....

> 5 yrs :

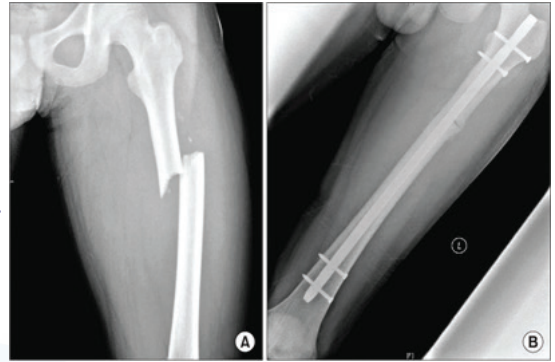
.....



Active space

After skeletal maturity :

.....

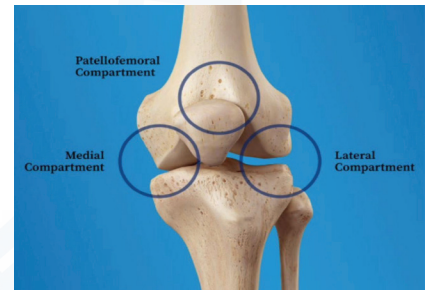


Complications:

- 1) Hypovolemic Shock
- 2) Fat Embolism Syndrome

Knee Anatomy

1. Hinge Joint
2. It is a tricompartmental joint:
 - i. Medial Tibio-femoral compartment :



- ii. Lateral Tibiofemoral Compartment

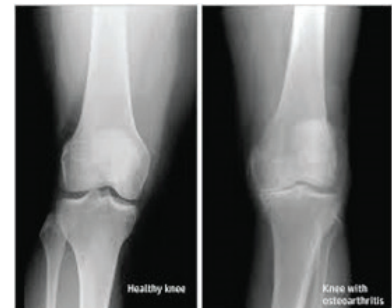
.....

.....

- iii. Patello Femoral Compartment

.....

.....



Knee Ligaments

1. Anterior Cruciate Ligament (ACL)

.....

2. Posterior Cruciate Ligament (PLL)

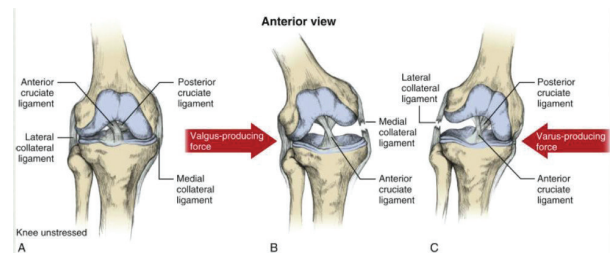
.....

3. Medial Collateral Ligament (MCL)

.....

4. Lateral Collateral Ligaments (LCL)

.....



- Both Collateral Ligaments are : &

- Both Cruciate Ligaments are : &

- Both Meniscus are : &

Knee Ligament Injuries:

Medial Collateral Ligament:

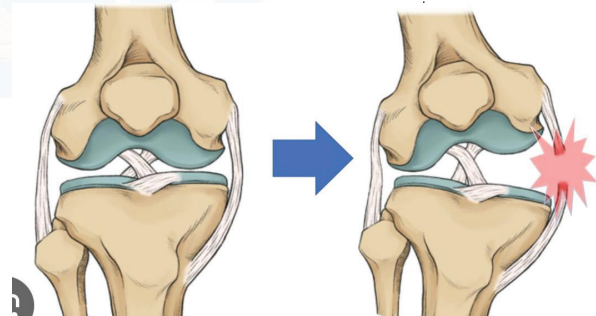
1. Injured due to

2. M/C Site of Injury :

.....

3. Clinical Test of Choice :

.....

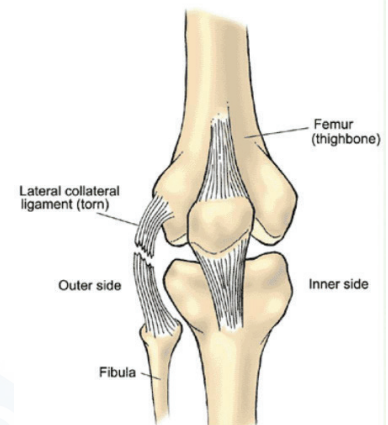


Lateral Collateral Ligament:

1. Injured due to

2. M/C Site of Injury :

3. Clinical Test of Choice :



Anterior Cruciate Ligament: (ACL)

i. M/C Ligament Injury in Knee Joint

ii. Cause:

iii. O'Donoghue's Triad : Injury to -

.....

iv. C/F:

v. Clinical Test:

i. Lachman's Test :-

.....



ii. Anterior Drawer Test: -

.....



iii. Pivot Shift Test :-

.....

IOC - MRI

Normal ACL



Torn ACL



Treatment:

- ⇒ Arthroscopic ACL reconstruction using
..... Tendon as graft



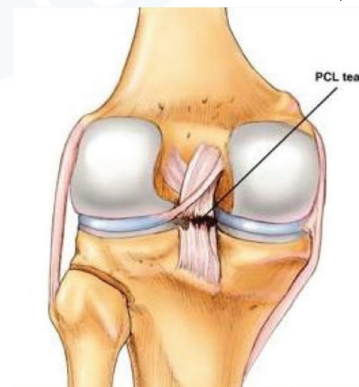
Posterior Cruciate Ligament:

- i. Difficulty in walking
- ii. Clinical Tests :

.....
.....

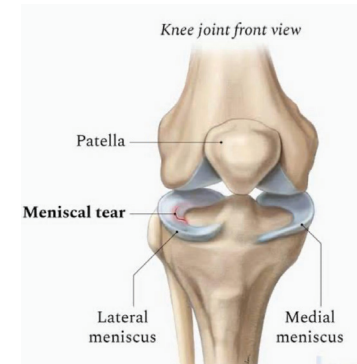
- iii. Treatment :

.....



Meniscal Injuries

1. It is a Fibrocartilaginous structure acting as shock absorbers
2. Consists of Type collagen
3. Blood Supply :
 - Medial 2/3rd
 - Lateral 1/3 rd
4. Mode of Injury
5. Injury to medial > Lateral meniscus (.....)
6. Pathoanatomy :
 - i) M/C Site of Tear :
.....
 - ii) M/C Type of Tear :
.....

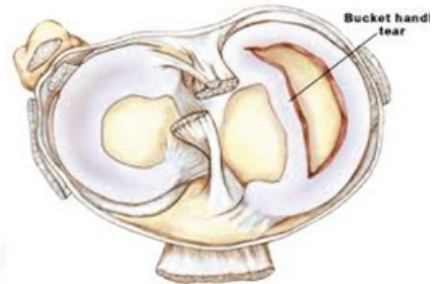


C/F :

.....

.....

.....

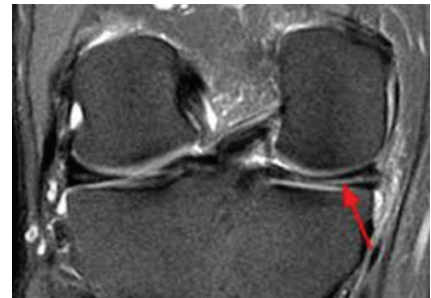


Clinical Test :

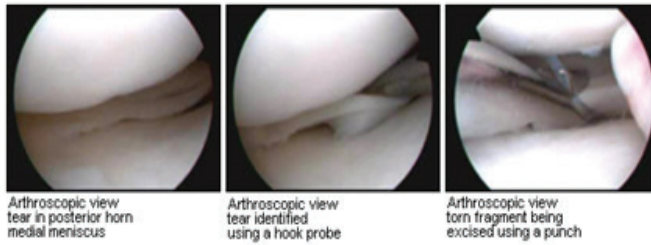
.....

.....

IOC - MRI



Treatment :



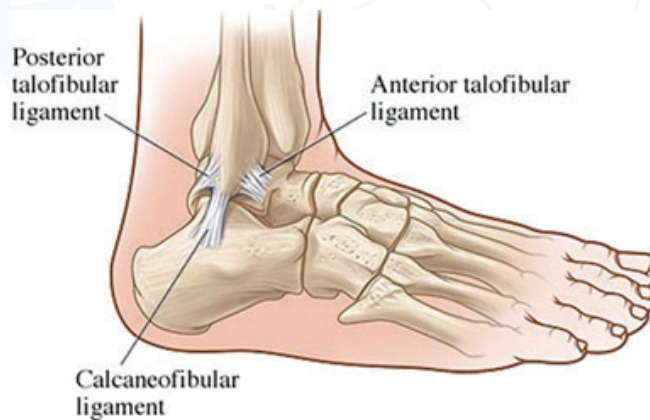
i. If tear is in outer 1/3rd (Vascular area) :

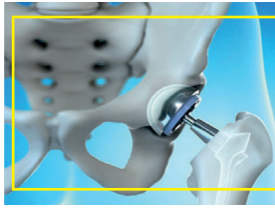
.....

ii. If tear is in medial 2/3rd (Avascular Area) :

.....

Injuries Around Ankle:

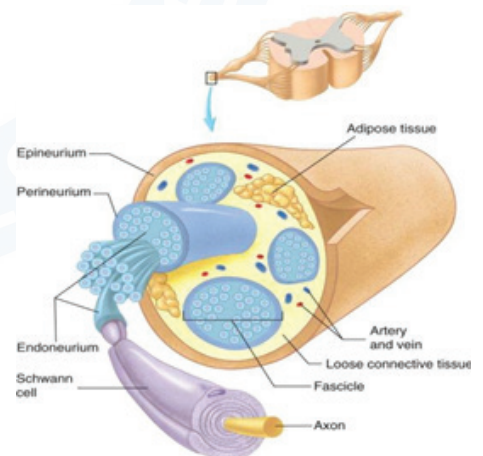
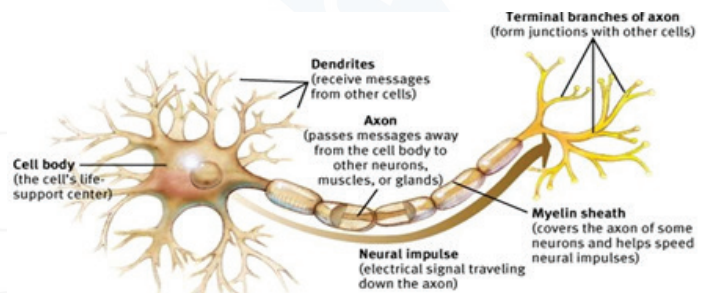




PERIPHERAL NERVE INJURIES

PERIPHERAL NERVE INJURIES

Peripheral Nerve Injuries



Pathology:

1. Wallerian / Secondary / Orthograde Degeneration: -

.....

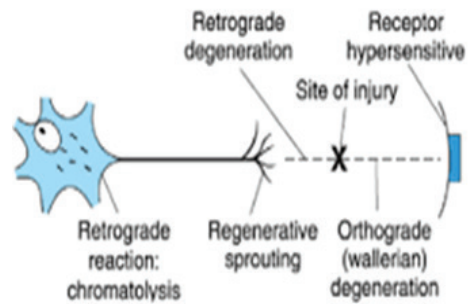
.....

2. Primary or Retrograde Degeneration: -

.....

.....

3. Total Time Required:

**Nerve Regeneration :**

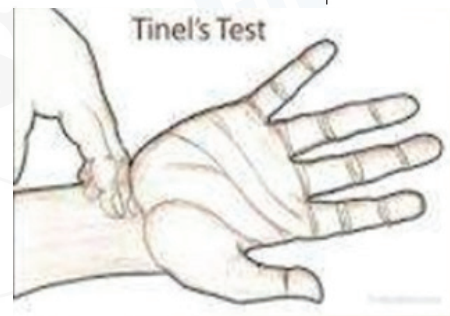
1. Proximal To Distal Direction

2. Rate :

3. Clinical Test :

.....

.....

**For Regenerating peripheral nerve :**

1. Advancing / progressive tincl's sign :-

.....

2. Motor March :-

.....

Best Test/Investigation to predict Nerve Recovery :

Active space

For Non - Regenerating nerve :

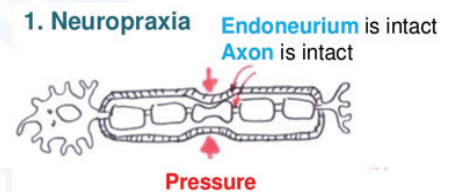
Classification of Peripheral Nerve Injuries :

I. Seddon's Classification :

3 types

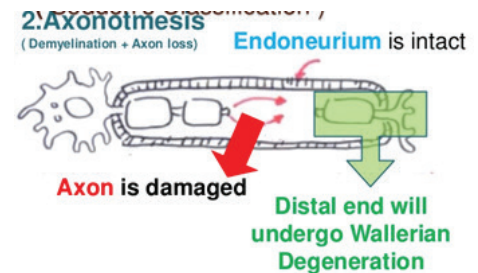
1 Neuropraxia:

- Physiological conduction block
- Nerve is anatomically intact
- Due to temporary disruption of
- Both wallerian & primary degeneration
- T/E : Spontaneous recovery in around 6 weeks



2 Axonotmesis:

- Injury to axon but outer endoneurium is intact
- Both wallerian & primary degeneration are
- Tinel's Sign :
- Advancing Tinel's Sign :
- Motor march :
- T/t : Surgical Intervention only if recovery stops



3 Neurotmesis: -

- Complete anatomical transection of nerve
- Both wallerian & primary degeneration are
- Tinel's Sign :
- Advancing Tinel's Sign :
- Motor march :
- T/t : Surgical Intervention.

Sunderland's Classification:

- Type - 1
- Type - 2, 3 & 4
- Type - 5

*Surgical Techniques for Nerve Injury:

1. Nerve Repair.

- Primary Repair
- Delayed Primary Repair
- Secondary Repair

2. Nerve Grafting :

.....

M/c Nerve Used : -

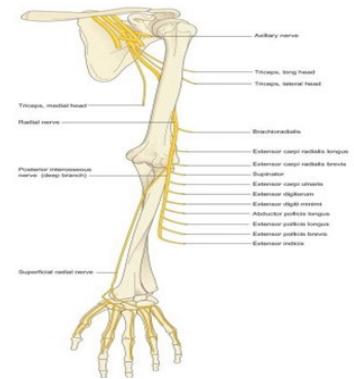
3. Tendon transfer surgery :

.....

.....

Radial Nerve

1. Branch of Posterior Cord of Brachial Plexus
2. Divides around Elbow into :
 - Sup. Sensory Nerve
 - Post Interosseous Nerve
3. Autonomous Zone:
(Exclusive Area of Sensory Supply By Nerve)
Dorsum of First Web Space



Motor Supply:

- 1 From Origin but Before Spiral Groove:
.....
- 2 Within Groove:
.....
- 3 After Groove & Before Division:
.....
- 4 After Division:
.....
.....

Types of Radial Nerve Palsies:

- 1 Very High Radial Nerve Palsy.
.....
.....

- 2 High Radial Nerve Palsy: (Injury In Spiral Groove)

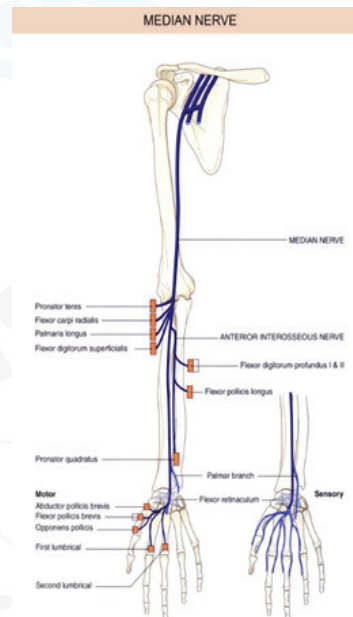
- M/C Cause :
- Leads to :
- Splint used :
- Tendon transfer :

3 Low Radial Nerve Palsy / Post Interosseous Nerve Palsy: (Injury of Nerve around Elbow)

Leads to
due to paralysis of

Median Nerve (Labourer's Nerve)

1. Is formed by joining of Medial Root and Lateral Root of the nerve
2. Gives Anterior Interosseus Nerve (Motor branch of deep anterior compartment muscles of forearm)
3. Autonomous Zone : Tip of Index Finger



Motor Branches:

In Forearm:

- Superficial Group :
- Deep Group:

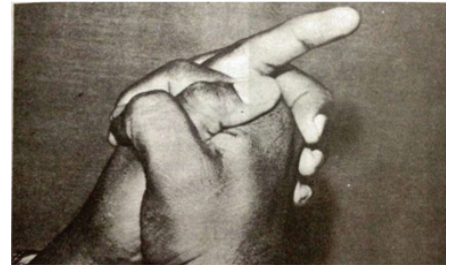
In Hand:

- Thenar Muscles:
- 1st & 2nd Lumbricals.

Clinical Test:

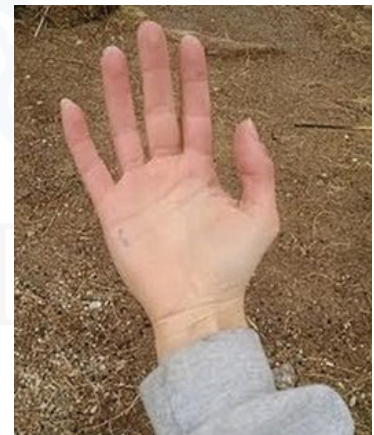
1. Pointing Index

Index finger remains on attempted fist/clasp formation due to paralysis of of Index finger .



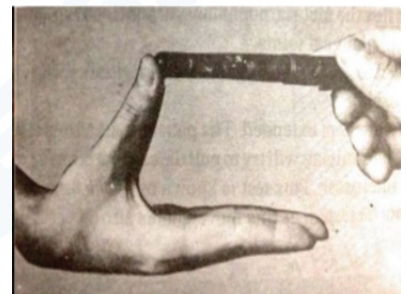
2. Ape Thumb / Simian Hand Deformity:

In paralysis of thumb is Extended and Adducted due to which it comes to lie in same plane as other fingers



3. Pen Test:

Inability to touch a pen kept at a distance due to paralysis of



4. Ok Sign / Kiloh - Nevin Test:

- (i) Ask patient to make a strong pinch between Thumb and Index finger
- (ii) Injury to nerve causes paralysis of &

FIGURE 4
Interosseous nerve syndrome:
A look at complete and incomplete paralysis

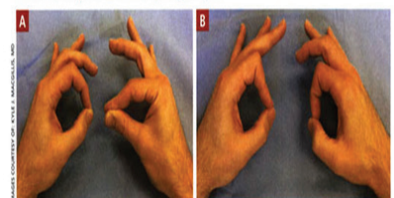


Figure 4A shows a right hand with complete flexor pollicis longus and flexor digitorum profundus index finger paralysis. Figure 4B reveals isolated flexor digitorum profundus index finger paralysis of the left hand. Both photos demonstrate evidence of anterior interosseous nerve syndrome.

Ulnar Nerve (Musician Nerve)

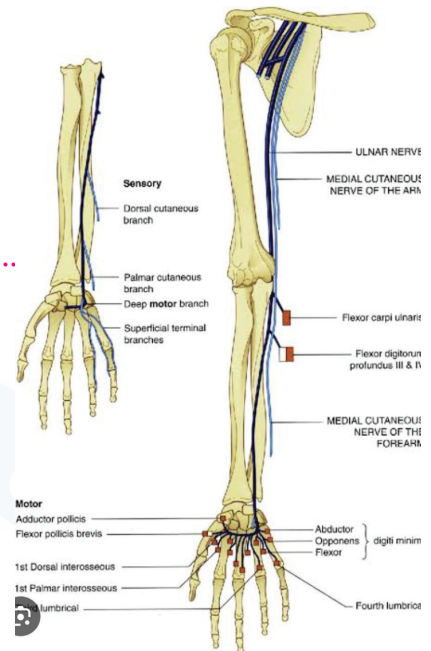
1. Branch of Medial Cord of Brachial Plexus.
2. Guyon's Canal :
3. Autonomous Zone :

Motor Branches:

1. Forearm:
.....
2. Hand:
.....
.....

Clinical Tests:

1. Claw Hand Deformity (Hyper extension of MCP joints and Flexion of IP joints)
 - i. Develops only in medial
2 fingers (.....)
 - ii. Occurs due to paralysis
of muscles
 - iii. Also called as
deformity



Active space

Types of Ulnar Nerve Palsy :

High Ulnar Nerve Palsy

- Injury To Nerve Proximal To Elbow
- All Muscles Are Paralyzed

Low Ulnar Nerve Palsy

- Injury Distal To Elbow
- Only Hand Muscles Are Paralyzed

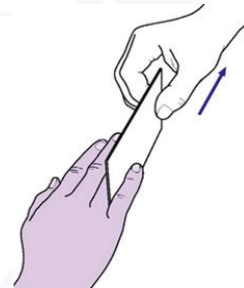
Ulnar Paradox:

Clawing is less severe in High palsy as compared to Low Palsy because of paralysis of

- Splint used :

2. Card Test:

.....



3. Egawa's Test :

Ask patient to move middle finger sideways



4. Book Test/Froment's Sign :

- (i) Ask patient to hold a book between thumb and other fingers
- (ii) In paralysis of patient holds the book by using as a substitute



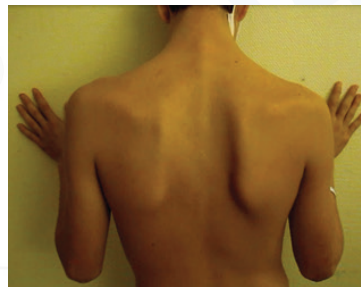
Long Thoracic Nerve (Bell's Nerve) - C5,6,7

.....

.....

.....

.....



Sciatic Nerve:

1. M/C Cause of Injury :

.....

.....

2. Leads to :

.....

.....

3. Splint used :

.....



Brachial Plexus Injuries:-

1. Erb's Palsy:

- Injury to trunk
- C5, C6 Root
- Supraclavicular Type of palsy
- Deformity:

.....

.....

.....



2. Klumpke's Palsy :

- Injury to trunk
- C8, T1
- Infraclavicular type of palsy
- Deformity :

.....

.....

.....

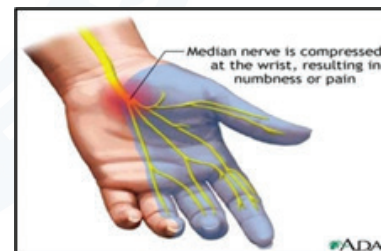


Carpal Tunnel Syndrome

1. M/C Entrapment Neuropathy.
2. There is compression of

.....

C/F : - 1. Pain, Numbness and Paresthesia in median Nerve distribution.



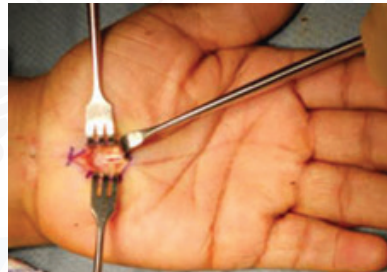
2. Wasting of muscles.



Clinical test: -



Treatment :



Active space

Tarsal Tunnel Syndrome:

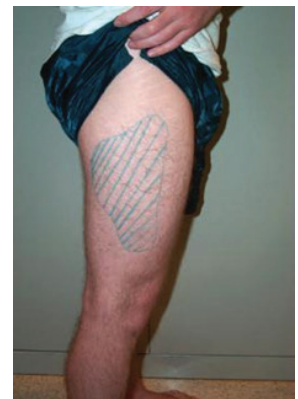
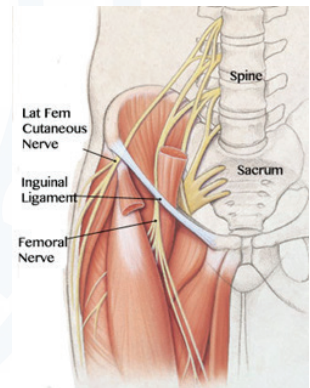
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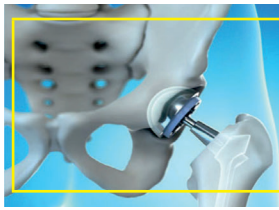
Ulnar Nerve Compression:

- 1) A round Elbow
-

- 2) Around wrist:
-

Meralgia Paresthetica :





INFECTIONS & TUBERCULOSIS

INFECTIONS & TUBERCULOSIS

Key points:

- 1) M/C Cause of musculoskeletal infection (Bone (or) joints) by pyogenic organism.

.....

- 2) M/C Cause of bacterial infection in

a) I/V Drug Abusers

b) Sexually Active Individual

c) In Haemoglobinopathies e.g Sickle Cell Anemia

.....

.....

- 3) M/C Cause of parasitic infection

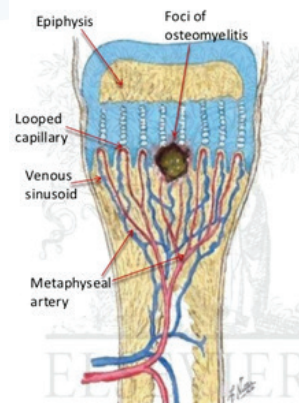
Acute Osteomyelitis

Duration:- < 2 wks

1. Definition :

.....

.....



Active space

2. Most common organism:

3. Mode of spread:

4. Most common site:

.....

Clinical features:

1. Pain, swelling
2. ↑ Local temperature
3. High grade fever
4. Decreased movements of affected limb

Investigations:

1) Blood tests:

.....

2) X rays:

1) Earliest findings on X-Rays (24-48 hrs)

.....

2) Earliest bony change on X-ray (7-10 days)

.....

3) Best Radiological test :

4) IOC:



Treatment:

1. If no pus is present: (24 - 48 hrs)

.....

.....

2. If Abscess is present: (More than 48 hours)

.....

Most Common Complication:

.....

Chronic Osteomyelitis (Duration > 3 months)

Occurs due to :

- (i) Delay in Onset of Treatment
- (ii) Inadequate duration of total antibiotic treatment
- (iii) Immunocompromised host / Increased Virulence of organism



Clinical Hallmark:

.....

X rays:

- 1) Dead, avascular bone surrounded by infected granulation tissue
 (Radiological Hallmark)
- 2) Radiodense new bone formation around sequestrum
- 3) Openings in bone for pus to come out



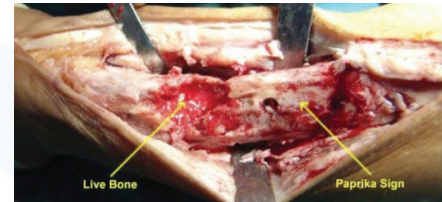
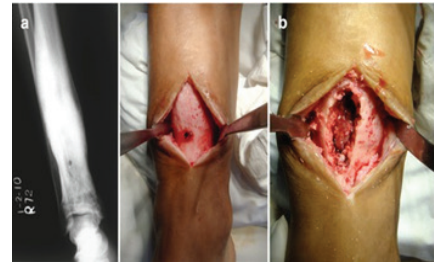
Treatment:**(i) Sequestrectomy :-**

.....

.....

(ii) Saucerization :-

.....

**Complications:**

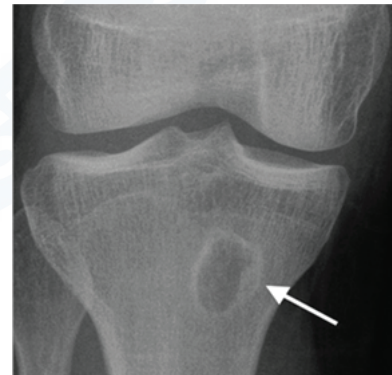
.....

Sub Acute Osteomyelitis

- Duration: 2 wks - 3 months
- Caused by : Organisms of Low virulence + Good host immunity

Brodie's abscess

- Age : -
- Site : -
- X-Ray :-



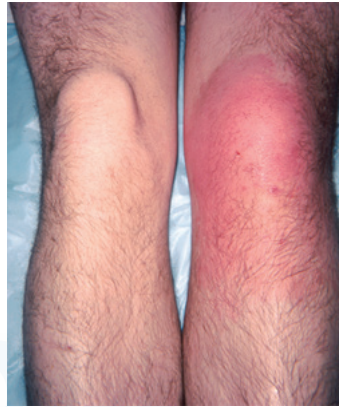
Septic Arthritis

1. Definition :

.....

2. M/C Cause :

.....



Most common Joint:

- Adult:

- Infants:

Mode:

- Investigation of Choice:

.....

Treatment:

.....

.....

.....

.....

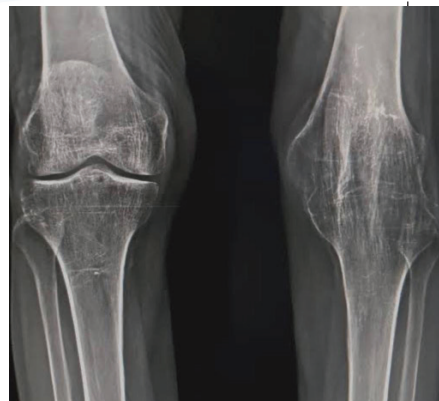
Complication:

.....

.....

.....

.....



Acute septic arthritis of HIP in infants:-



Infection of skin & subcutaneous tissue in foot by :

- i. Actinomycetes
- ii. Eumycetes

C/F :

.....

.....



Musculoskeletal Tuberculosis

Pott's Spine:

1. M/C Site of Bone and Joint tuberculosis (> 50 % cases)

2. M/C Site :

- i. In adults :
- ii. In children :

3. Route of spread : (Haematogenous)

Direct arterial

Paravertebral venous plexus

(.....)



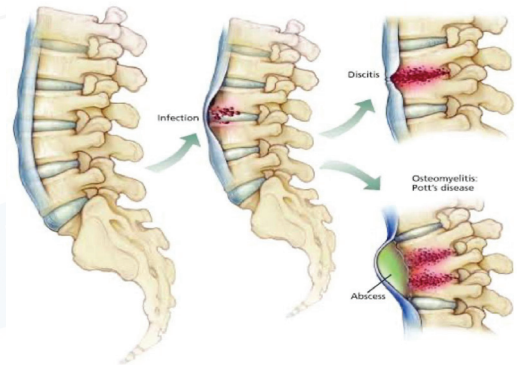
4. M/C Part of Vertebrae Involved :

.....

5. Pattern of Involvement:

Involvement of 2 adjacent vertebrae and
Intervertebral disc in between :

.....



Clinical Features:

1 M/C and Earliest symptom is :

2. Earliest Sign :

Active space

3. Cold Abscess :

.....

4. Pott's Paraplegia (20 %) :

.....



X Rays:

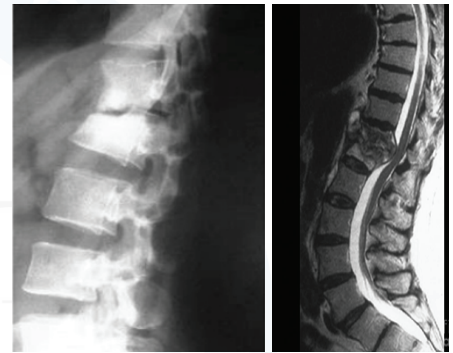
Earliest X Ray sign :

.....

.....

Investigation of Choice :

.....



Treatment :

1. Middle path regime :

.....

- If Clinical Improvement is present :

.....

- If Clinical Improvement is not present :

.....

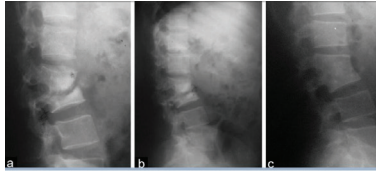
2. Treatment of Cold Abscess :

.....

.....

3. Sequelae of Spinal Tuberculosis :

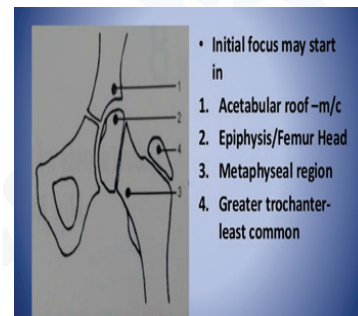
.....



TB HIP :-

- (i) Second most common site of
 Musculoskeletal Tb, but
 M/C site of

- (ii) M/c initial focus of Infection :



Stages of TB HIP :

- (i) Stage of Synovitis :

.....

- (ii) Stage of Early arthritis :

.....

- (iii) Stage of Advanced Arthritis :

.....

- (iv) Stage of advanced arthritis with joint subluxation and
 dislocation :



.....



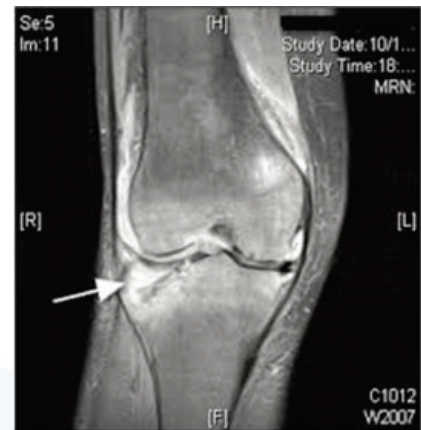
.....

TB Knee :-

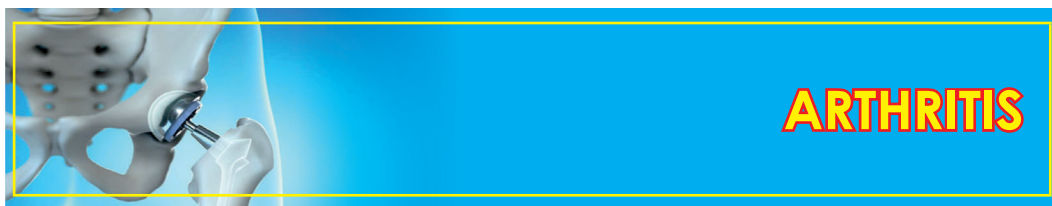
.....

.....

.....



Active space



ARTHRITIS

1. Definition : Pain, Swelling and Limitation of joint movement

2. Inflammatory & Non -Inflammatory arthritis:

i. Inflammatory Arthritis :

- Seropositive :

.....

- Seronegative :

.....

ii. Non Inflammatory Arthritis :

.....

Rheumatoid Arthritis:

1. M/C form of Inflammatory Arthritis (.....)
2. More common in
3. It is an disease
4. It is bilaterally arthritis of small joints of hand and foot
5. Morning stiffness

Aetiopathology:

Genetic Predisposition

.....



Ag - Ab complexes (.....)



.....



.....

Joints Involved:

- M/C joint involved :

- are spared in Rheumatoid Arthritis

- M/C Large Joint Involvement :

- M/C Spine Involvement :

Deformities in RA:

I. Hand Deformities :

1. Z- deformity:

.....

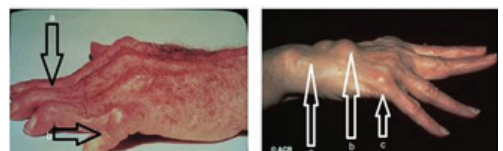
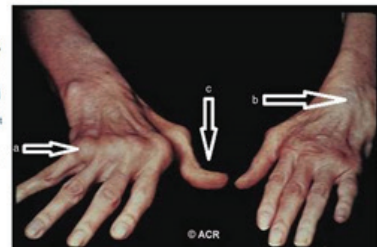
.....

2. Swan neck deformity:

.....

.....

Rheumatoid Arthritis, Clinical Features,
Fig. 3 Prominent ulnar deviation in the right hand, metacarpophalangeal (MCPs) (a), and proximal interphalangeal (PIP) swelling in both hands and synovitis of left wrist (b) and Z-deformity of the right thumb (c) (© 2013 American College of Rheumatology. Used with permission)



3. Boutonniere's deformity:

.....

.....

II. Knee Deformities :

.....

.....

.....

X Rays :



Blood Test:

1. RA Factor :

.....

.....

2. Anti CCP antibodies :

.....

.....

Treatment:

1. DMARD's :

.....

.....

2. Steroids :

.....

.....

3. Biological Therapy :

.....

.....

.....

Ankylosing Spondylitis:

1. Sero Negative Spondylo Arthropathy
2. Disease → Inflammation of Spine →
3. Involves Axial skeleton (.....)
4. More common in
5. M/C and First joint involved :
6. Patient is positive for

Pathology:

Immune Complexes Deposition



.....



Cartilage Destruction



.....

Normal
posture



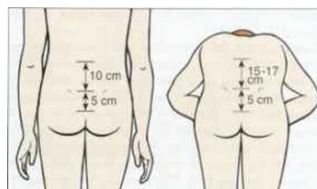
Advanced
ankylosing
spondylitis



Clinical Test:

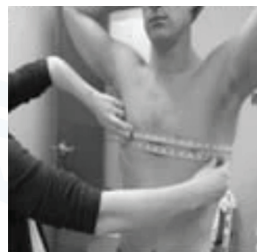
1. Decreased Flexion at Lumbar Spine :

.....



2. Chest Expansion test :

.....



3. SI joint tests :

.....

X-Rays:

1. 1st X-Ray finding :

.....



2. X Ray Spine Lateral view :

.....



3. X Ray Spine AP view - Central Radiodense Line (Calcification of Supraspinous and Interspinous Ligaments) :

.....



Active space

T/T :

- 1) NSAID's -
- 2) Physical Therapy
- 3) DMARD's, Steroids, Biological Therapy

Reiter's Disease: (Reactive Arthritis)

Seronegative

Triad of :

.....

.....

.....

Psoriatic Arthropathy:

1. It is a Seronegative poly arthritis
2. M/C joint involved :



.....

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Treatment :

.....

Crystal Arthropathy: -

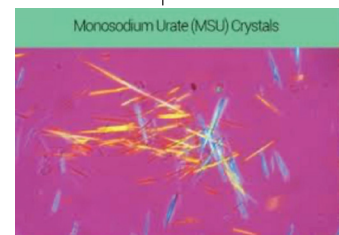
Gout:

- i. Occurs due to
-
-
- ii. Deposition of crystals of
-
- iii. M/C joint involved :
- iv . Bursitis : M/ C bursa involved
-



Diagnosis :

- 1) It is a clinical diagnosis
 - 2) Synovial Fluid Analysis under polarised microscope :
-



Treatment :

1) Acute Gout :

.....

.....

2) Chronic Gout :

.....

.....

Pseudogout:

1. Occurs due to deposition of crystals of

2. M/C Cause :

3. M/C joint involved :

Pseudogout + Alkaptonuria leads to :

.....

.....

**Diagnosis :**

Synovial Fluid Analysis under polarised microscope :

.....

.....

Calcium Pyrophosphate Dihydrate (CPPD) Crystals



Treatment :

Osteoarthritis:

Definition : Age related degeneration of articular cartilage

It is of 2 types :

i. Primary OA : (No pre-existing disease)

Cause :

M/C joint involved :

.....



ii Secondary OA : (Due to an underlying pathology)

M/C joint involved :

C/F:

1. Earliest and M/C symptom :
2. M/C muscle wasting seen :

Active space

3. In Hand OA :

i M/C joint involved :

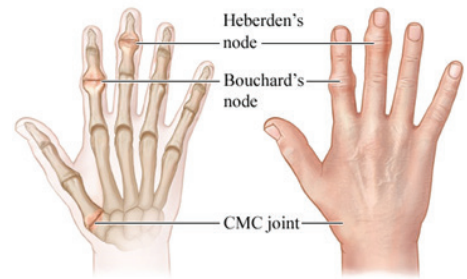
.....

ii OA of PIP joints :

.....

iii OA of DIP joints :

.....



X Rays :

- 1) Decreased medial knee joint space
- 2) Subchondral Sclerosis
- 3) Osteophytes and Loose body formation



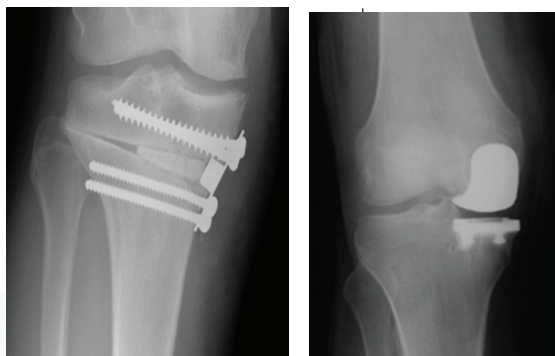
Treatment:

1. Conservative :

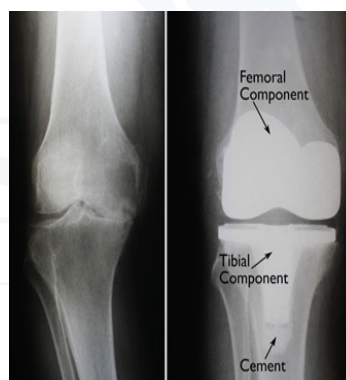
- i) NSAID's - Pain Relief
- ii) Physical Therapy :
- iii) Intra - articular injections of

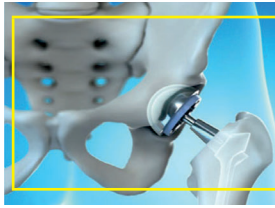
2. Surgical Procedures :

- i. Young patients with only medial compartment OA :



- ii. Elderly patients with Advanced OA :





TUMORS

TUMORS

Key Points:

1. M/C Malignant Bone Tumor :
2. M/C Primary Bone Malignancy :
3. M/C Benign Bone Tumor :
4. M/C True Benign Bone Tumor :
5. M/C Bone Tumor in Hand :

Metastasis / Secondaries :

M/C Malignant Bone Tumour :

Types :

1. Osteolytic →
2. Osteoblastic →
3. Mixed →

Sites : Spine (M/C)

M/C Cause of Metastatic Bone Tumour :

1. In Males →
2. In Females →

Classification of Tumors (WHO)

1. Bone Forming Tumors:

Benign :

Malignant :

2. Cartilage Forming Tumors:

Benign :

Malignant :

3. Giant Cell Tumor (Osteoclastoma)

4. Marrow Tumors

.....

.....

5. Bone Cyst

.....

.....

Distribution Of Bone Tumors:

1. Epiphyseal Tumors :

.....

.....

2. Metaphyseal Tumors:

.....

.....

.....

.....

3. Diaphyseal Tumors: -

.....

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.....

.....

Osteoid Osteoma:

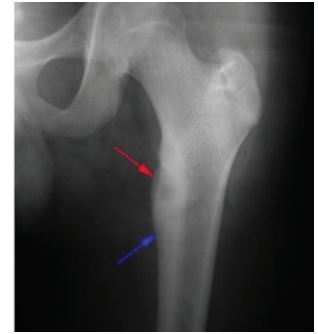
1. M/C "True "Benign Bone Tumor

2. Age group :

3. Site :

Active space

4. Location :

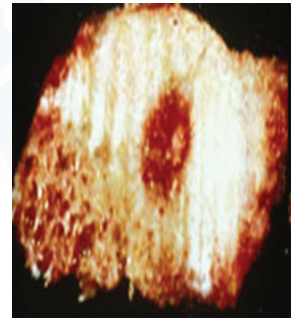


C/F:

.....

Pathological lesion :

.....



Treatment:

.....

Osteosarcoma:

1. Malignant Bone (Osteoid) forming tumour

2. Types:

i Primary Osteosarcoma : (No premalignant lesion)
 Age : 15-20 yrs

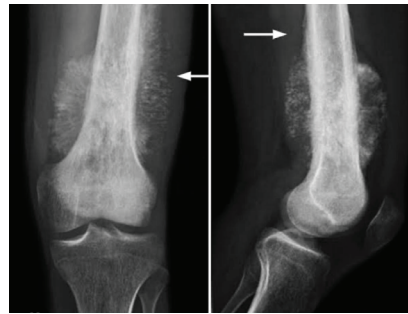
ii Secondary Osteosarcoma :
 (Predisposing condition is present)

.....

3. Site :

.....

.....



4. Location :

.....

.....

5. Genetic Association :

.....

.....

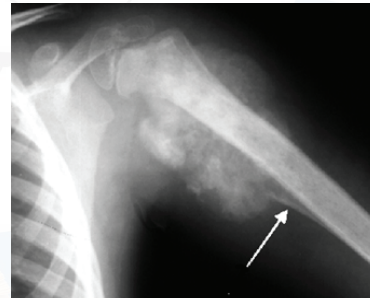
6. Metastasis:

.....

7. X Rays :

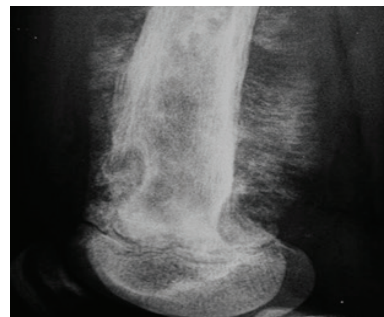
Triangular area of Subperiosteal
New Bone formation :

.....



Tumour growth in soft tissues is along
the blood vessels :

.....



Active space

Treatment :

1. Osteosarcoma is atumour

T/t : Chemotherapy & Surgery

Pre-operative Chemotherapy (.....)



Surgery



Post - operative Chemotherapy (.....)

Osteochondroma / Exostosis:

1. M/C Benign Bone Tumor
2. Age group :
3. Site :
4. Location :
5. Types :
.....
6. Pattern of growth :
.....
7. M/C Complication :
8. Treatment :
.....



Enchondroma :

1. M/C Bone tumour of Hand
2. Age group :
3. Site :
4. Location :



5. Multiple Enchondromatosis :

.....

.....

6.
-
-



7. Treatment :

.....

Chondrosarcoma :

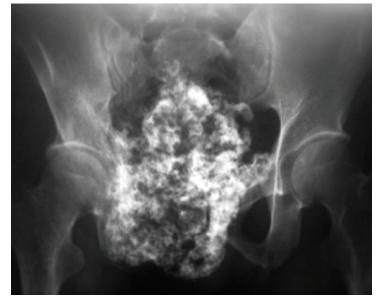
1. Malignant Cartilage forming tumour
2. Age group :
3. Site :

4. Location :

5. X Rays :

.....

.....



6. Treatment :

.....

.....

Giant Cell Tumor:

1. Also called as :

2. Benign but locally aggressive tumour

(.....)

3. Age group :

4. Site :

.....

5. Location :



6. X Rays :

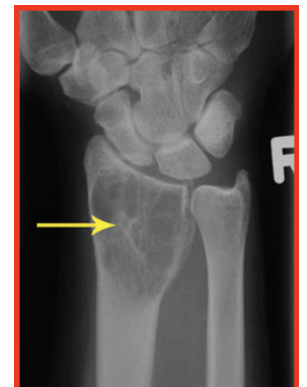
.....

.....

7. Treatment :

.....

.....



Fibrous Dysplasia;

1. Normal Bone and marrow is replaced by
2. Age group :
3. Site :

.....

4. Location :

5. Hormonal Association :

(Polyostotic Fibrous Dysplasia,

Precocious Puberty &

Skin pigmentation)

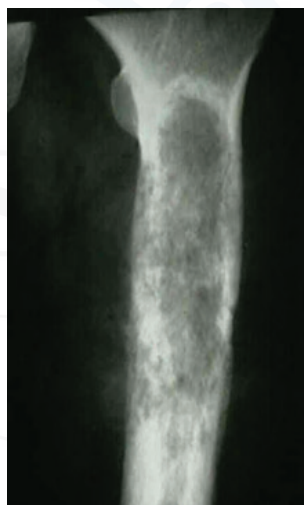


.....

6. X-Ray:

.....

.....



7. Treatment :

.....

.....

Simple Bone Cyst & Aneurysmal Bone Cyst

I) Simple Bone Cyst :

- Age :-
- M/c Site :-
- Cyst Consists of inflammatory fluid



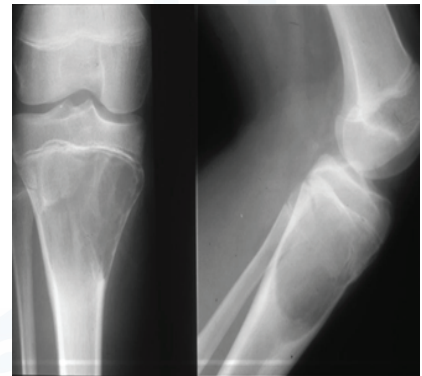
- X-Ray :-

.....

.....

II) Aneurysmal bone cyst :-

- Age :-
- M/c Site :-
- ABC is a blood filled cavity
- X-Ray :- Eccentric Metaphyseal lesion



Ewing's Sarcoma

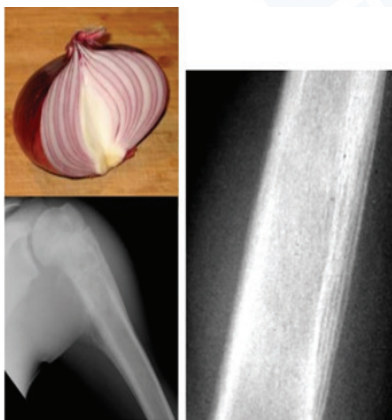
- Blue round cell tumour
- Age :-
- Location :-
- Site :-
- C/F :- Mimics S/S of

.....



- X-Ray :-

.....



- Cytogenetics :-

M/C

- T/t :-

.....

Active space

Multiple Myeloma

- M/C Primary bone malignancy
- Tumour of Cells
- Age :-
- Site & Location :-

.....

.....

- C/F:-

.....

.....

- Investigations

1 Blood :-

⇒ ↑ ESR

⇒ ↑ S. Calcium but normal ALP

⇒ LFT - A : G → 1 : 2 (Reversal of ratio)

⇒ Serum Electrophoresis (M-Spike)

2 Urine :-

⇒ Light chain immunoglobulin

In Urine

3 Investigation of choice :-

.....

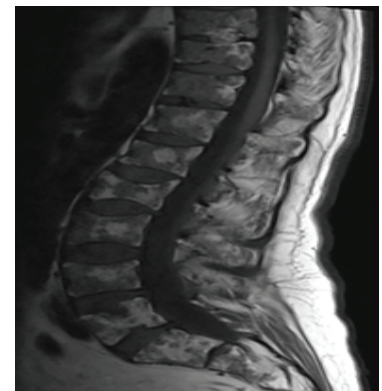
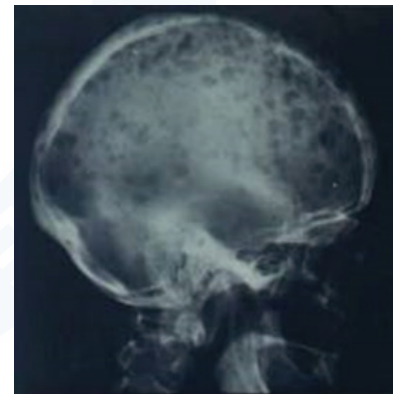
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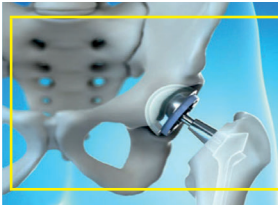
- X-Rays

.....

.....

T/t -





CONGENITAL DISORDERS

CONGENITAL DISORDERS

Congenital Disorders

Congenital Talipes Equinovarus (CTEV)

M/c Cause :

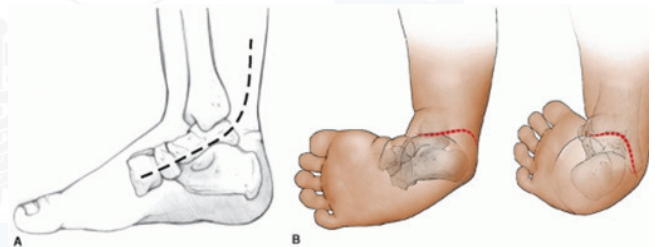
Deformity Components: (CAVE)

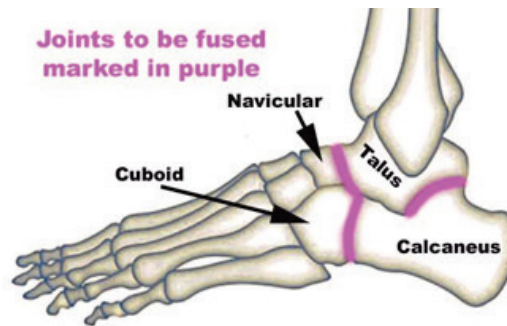


Active space

X-rays:

T/t:

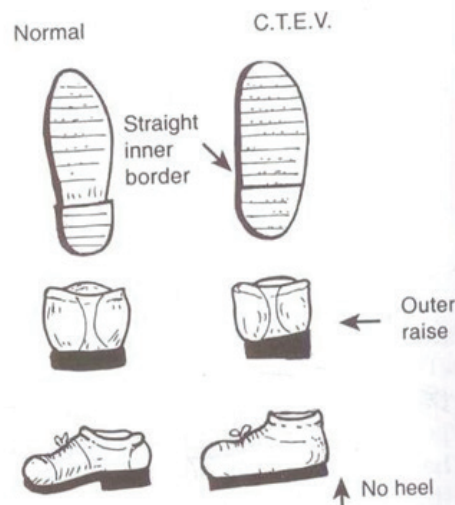




Dennis Brown Splint

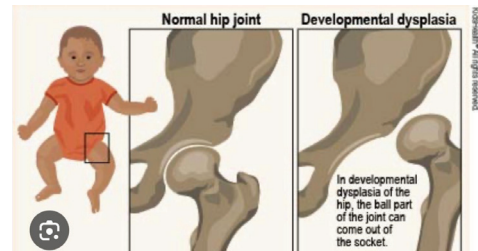


CTEV Shoes:



Active space

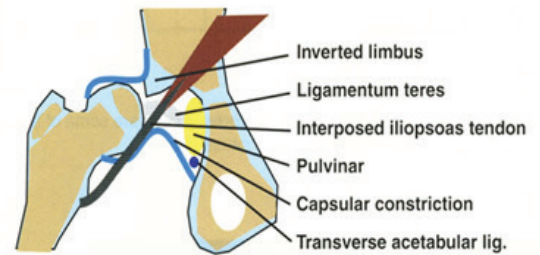
Developmental Dysplasia of Hip / Congenital Dislocation of Hip



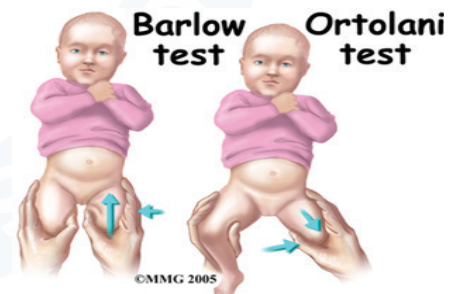
Cause:

1. Females > Males
2. Breech Malposition
3. Heredity

Pathoanatomy:



Diagnosis



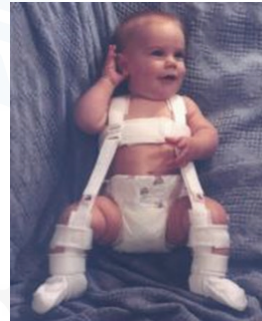
Xrays:



T/t:



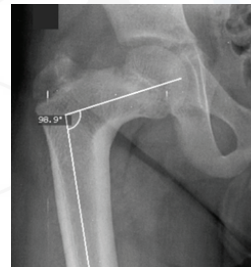
Splints used are:



Congenital Coxa Vara:

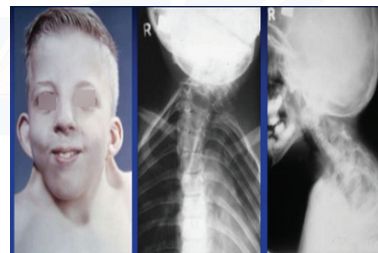
1. Reduced Neck - Shaft angle (< 120 deg)
from Birth

2. Restriction in &
.....



Klippel-Feil Syndrome:

1. Congenital fusion of
..... vertebrae

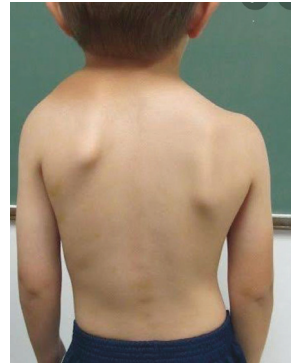


2. Triad of :

.....
.....
.....

Active space

Sprengel's Shoulder



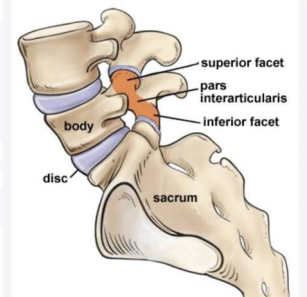
Spondylolysis :

1. Break in



2. M/c Site of Spondylolysis :

.....



3. X Ray view of choice :

.....

